She Code Africa Academy

Data Analysis

Price Prediction of Avocado

Presented by: **Group B SCA Data Science Track**

Project Overview

This presentation provides an overview of the predictive data analysis process of the price of Avocado, focusing on how data can be explored by predictive models. The goal is to understand the structure of the data, identify meaningful patterns, and present findings in a clear and informative way.

Objectives

Apply Linear Regression and Time Series models for prediction

Apply Logistic Regression and PCA for classification

Perform data cleaning and feature engineering

Evaluate and interpret machine learning models

Dataset Description

The dataset contains structured information collected over a period of 3 years(2015-2018). It includes the average price of avocado, the types of avocado, and the total volume purchased per time. Each variable provides valuable information that contributes to understanding the patterns and relationships for the overall analysis.

Data Preparation

This step involves interrogating the dataset, cleaning the data, removing missing data, and ensuring that the information is ready for analysis. Proper preparation improves the quality and reliability of the results.

Interrogating the Data & Data Cleaning

In the dataset, there were no missing values found.

```
# Checking for duplicates
avocado_duplic = avocado.duplicated().sum()
print(f'The number of duplicates records are: {avocado_duplic}') #No duplicates

#Checking for missing values
avocado_missing = avocado.isna().sum()
print(f'The number of missing values in each columns are: \n{avocado_missing}')

#Cleaning the column names of the data
avocado1 = avocado.clean_names()
avocado1 = avocado1.rename(columns={'averageprice': 'average_price'})

#Dropping unnecessary column 'unnamed_0'
avocado1 = avocado1.drop('unnamed_0', axis = 1)
avocado1
```

Data Preparation

Results of Data Cleaning & Interrogation

- The data was collected from 54 regions in the United States.

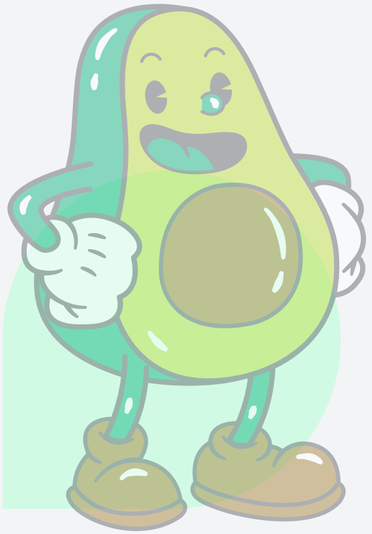
- The dataset contained 14 columns and 18249 records with no missing and duplicate entries/values.
- The date column was converted to datetime format.

- The column names were cleaned using the .clean_names() function in janitor and the unnecessary 'unnamed:0' column was dropped.

- All the numerical columns in this dataset are positively skewed (right skewed). The xlarge_bags and 4770 are the highest skewed. The average price slightly skewed (almost normally distributed).

Exploration Process

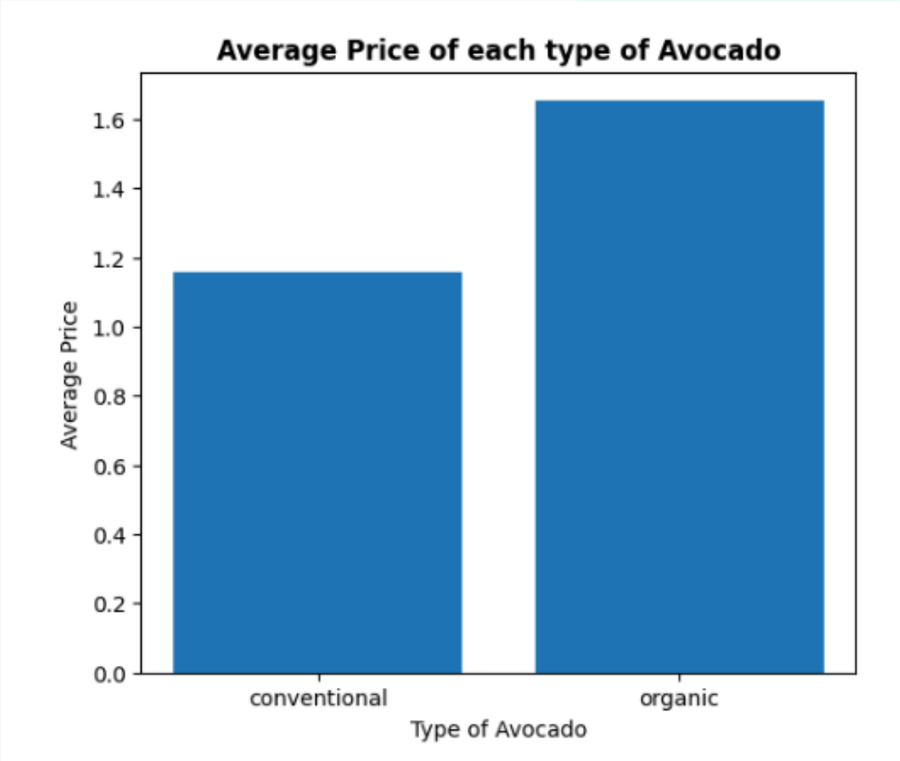
The dataset is explored to examine and understand its characteristics, in order to identify patterns, check distributions, and observe relationships between variables. Exploration helps build a foundation for deeper analysis.

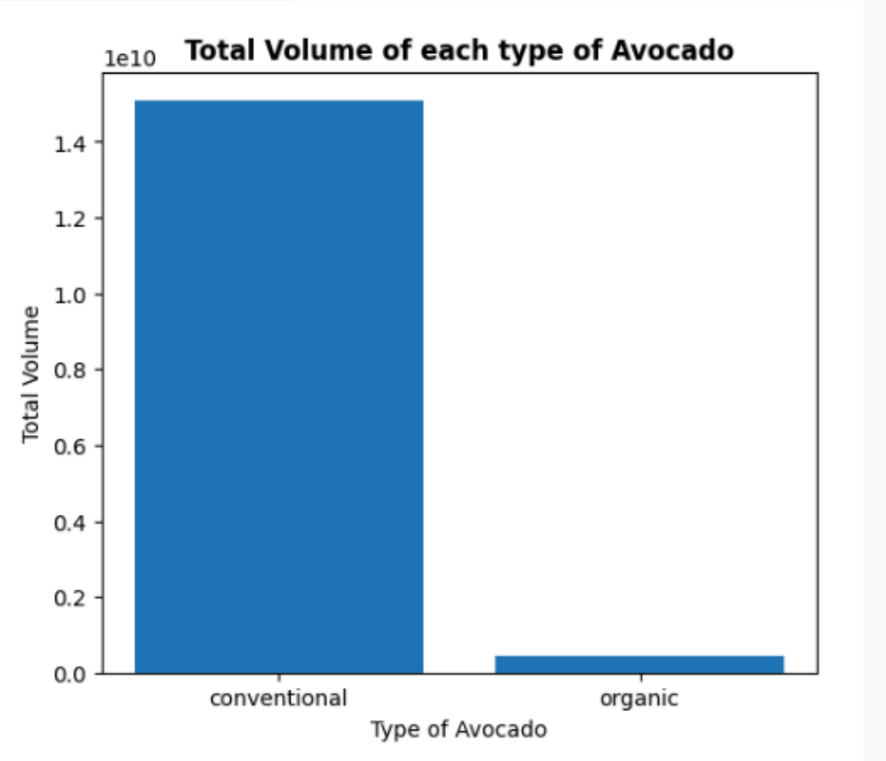


Let’s explore the dataset!

Data Exploration

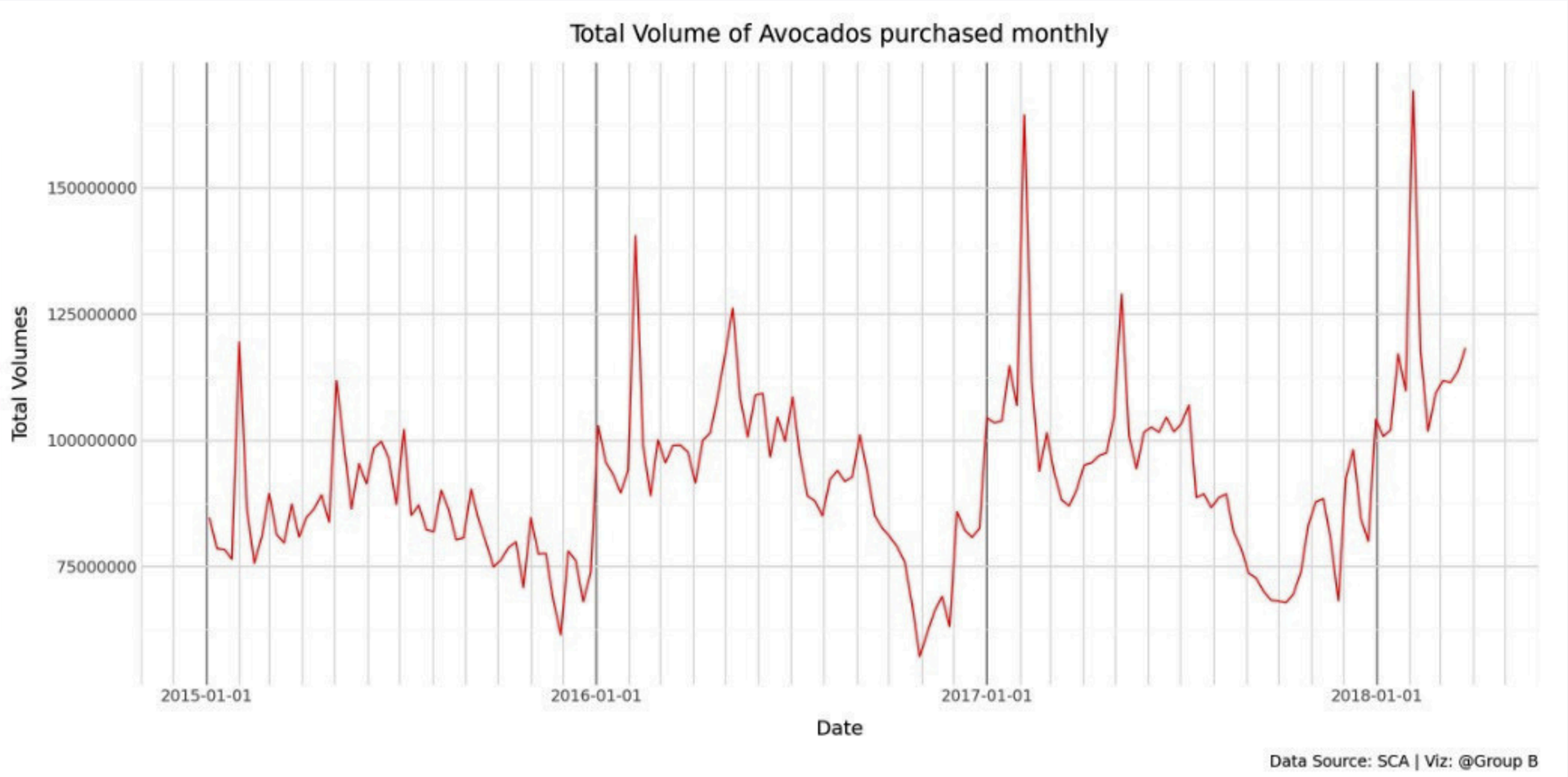
The bar chart depicts the difference in the average price of the different types of avocado.
Accorgind to the plot, organic types of cost more than the conventional type.





Data Exploration

We took a step further to plot a visual representation of the monthly consumption of avocados, and observed a recurring spike in the month of May.

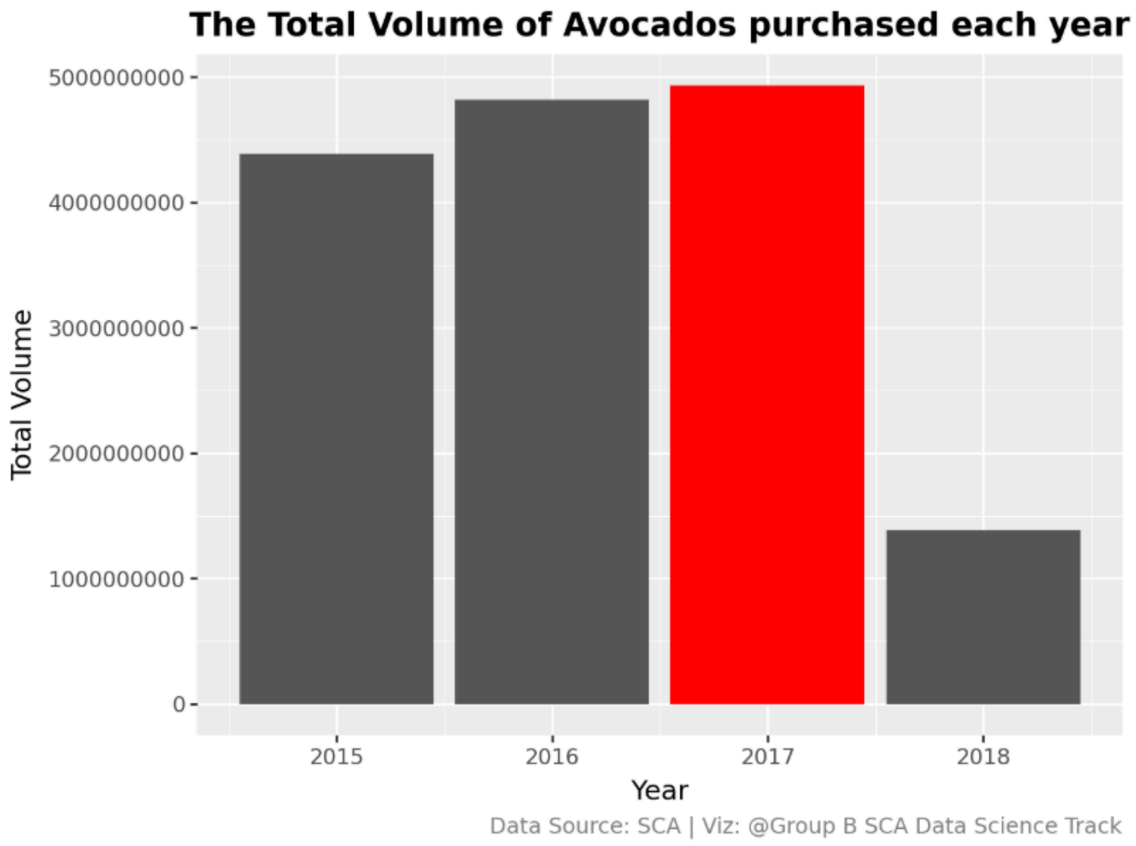


Data Exploration

This graph shows the year with the highest volume of purchased avocado.

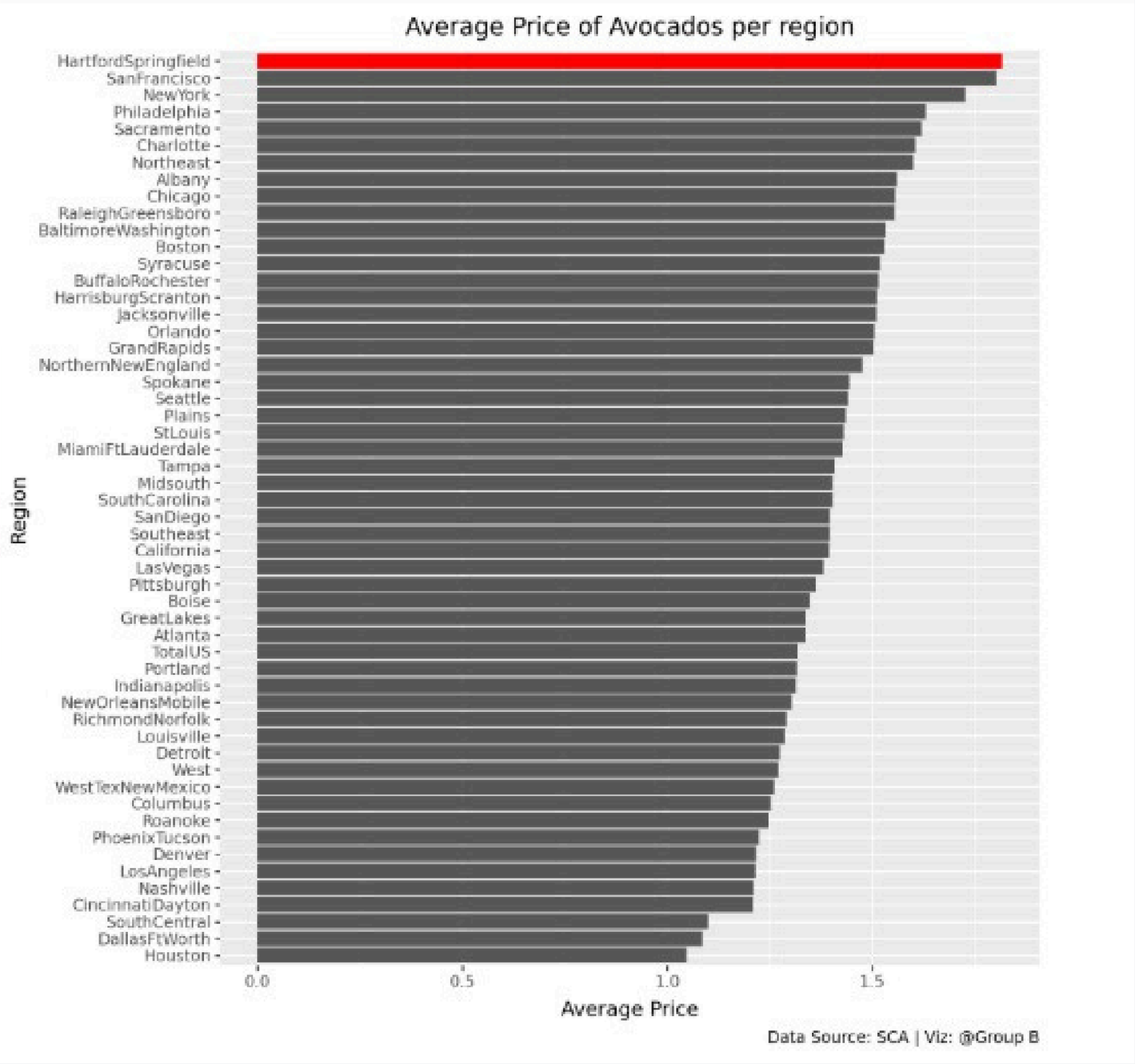
```
[20]: (ggplot(avo_year_vol, aes('year', 'total_volume')) + geom_col() + geom_col(av_2017, aes('year', 'total_volume'), fill = 'red
 + labs(x = 'Year',
        y = 'Total Volume',
        title= 'The Total Volume of Avocados purchased each year',
        caption= 'Data Source: SCA | Viz: @Group B SCA Data Science Track')
+ theme(plot_title=element_text(weight='bold'), plot_caption=element_text(color='grey')))
```

[20]:



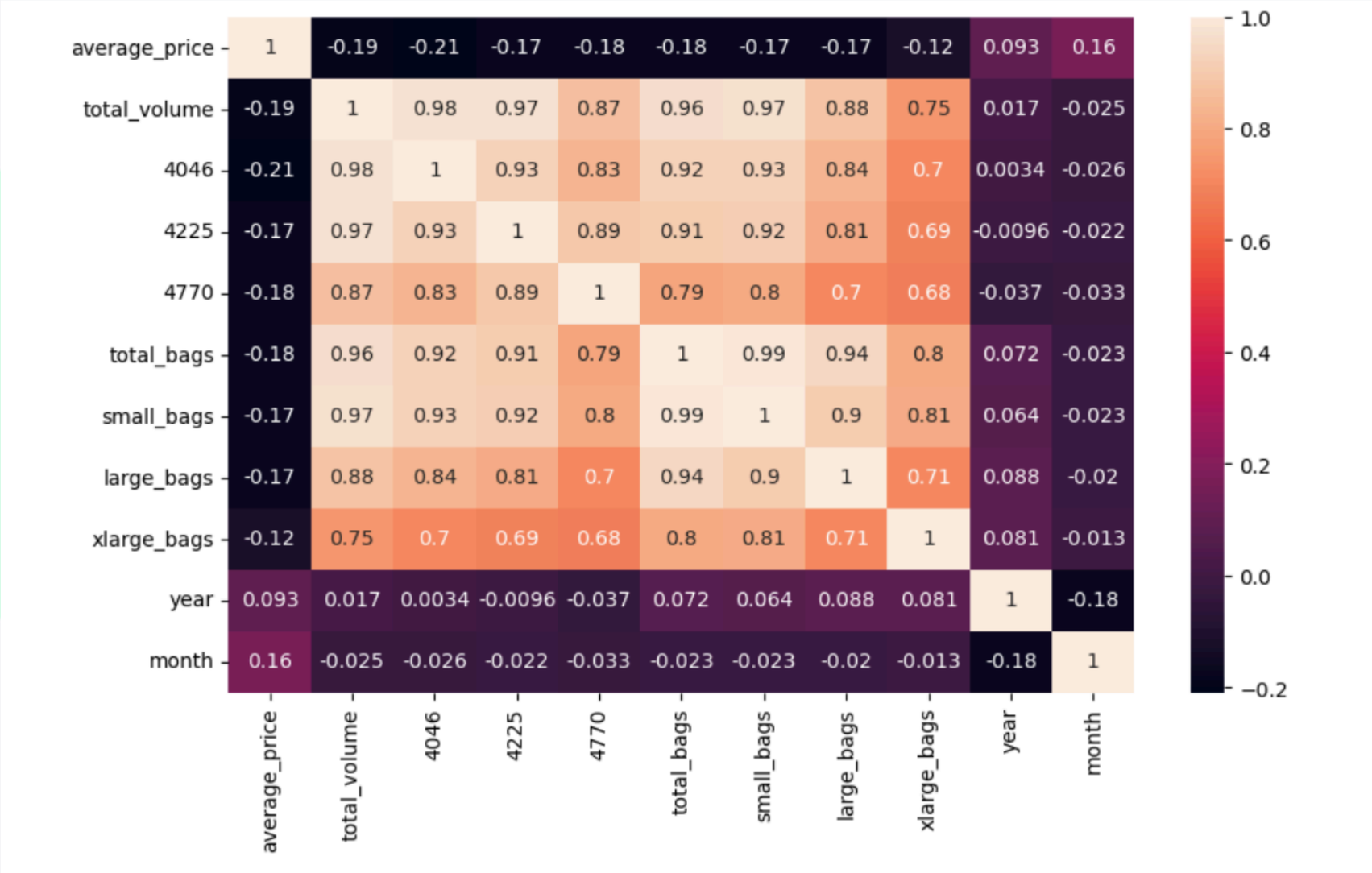
Data Exploration

This graph shows the average price of avocados in those regions.
Avocados are most expensive in the Hartford-Springfield region and least expensive in the Houston region.



Data Exploration

This shows the highest and least correlated feature to average price of avocados. The highest correlated feature to Average price is 4046: -0.21
The least correlated feature to Average price is year: 0.09

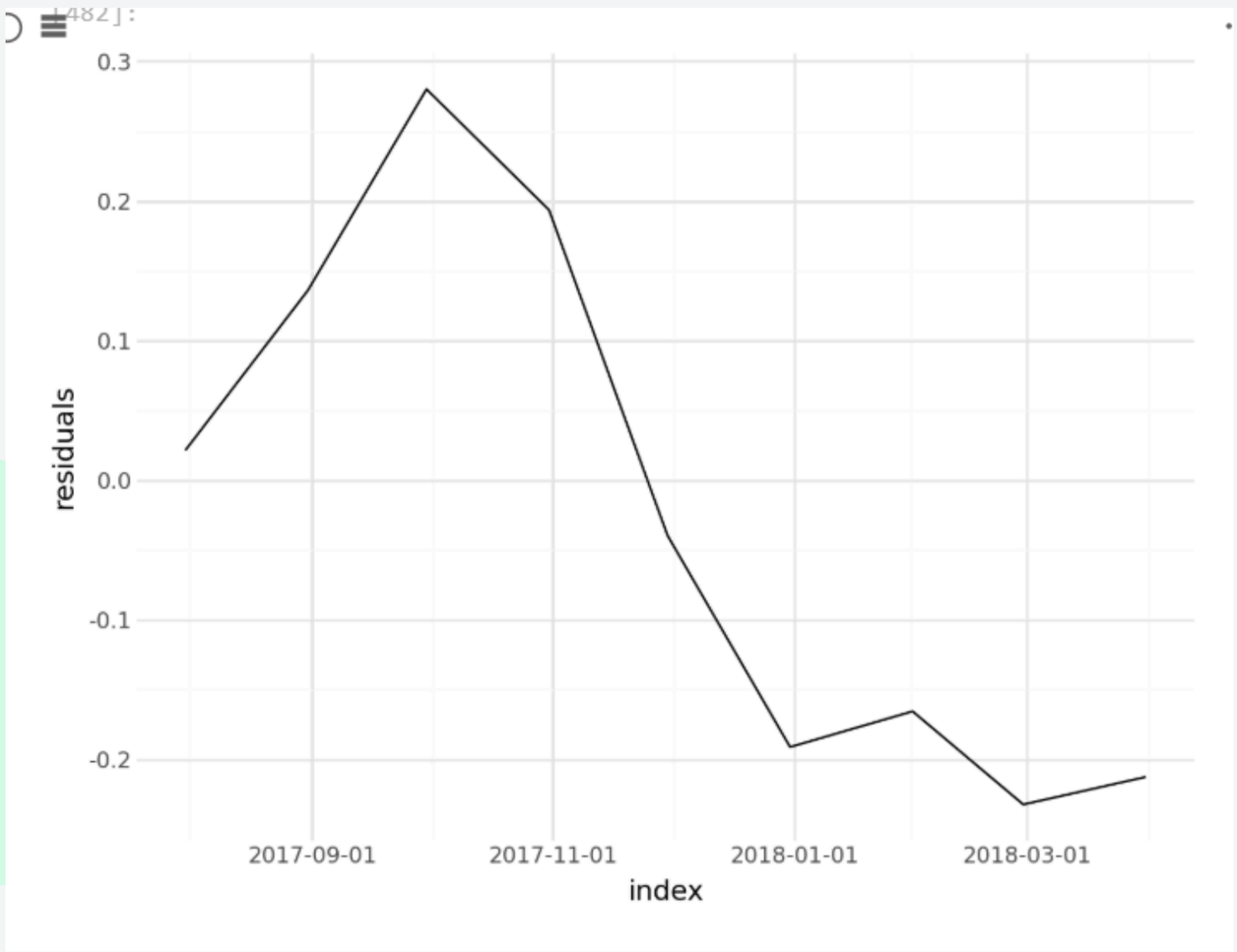


Data Exploration

Results of Data Exploration

- Two avocado types: Organic (premium-priced, low volume) vs. Conventional (affordable, high volume) — type is a core price driver
- 2017 was the peak year for both price and sales volume, likely driven by reduced harvest yield from poor weather or pest activity, yet sustained high demand
- Demand spikes in 3 key months:February — Super Bowl (guacamole)
- May — Cinco de Mayo (Mexican cuisine)
- July — Independence Day & National Avocado Day (July 31)
- Region drives price: Hartford-Springfield is the most expensive region due to long supply chains from harvest zones; Houston is the cheapest, bordering Mexico — a key harvest region
- Harvest regions (California, Mexico, Florida) — proximity reduces cost; the West and TotalUS lead in purchase volumes, boosted by California's local supply
- Top price predictors: Avocado type · Season/Month · Region · Year · Supply-side disruptions.

Observed Trends



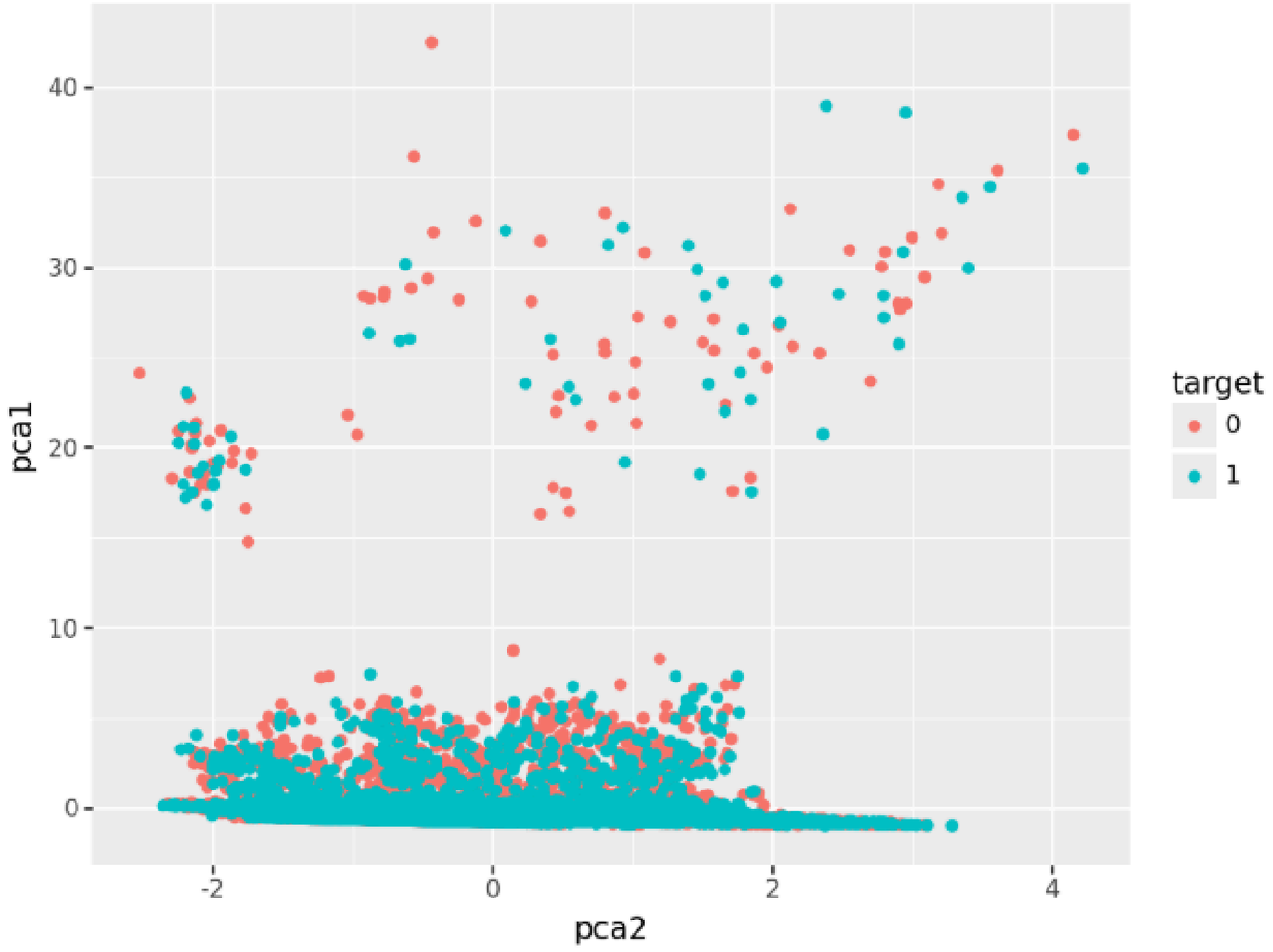
Key drivers of avocado prices

- The analysis shows that time (seasonality), the type of avocado and the total volume are the key drivers of avocado prices.

Analytical Approach

Avocado Type Classification using PCA and Logistic Regression Classification

[37]:



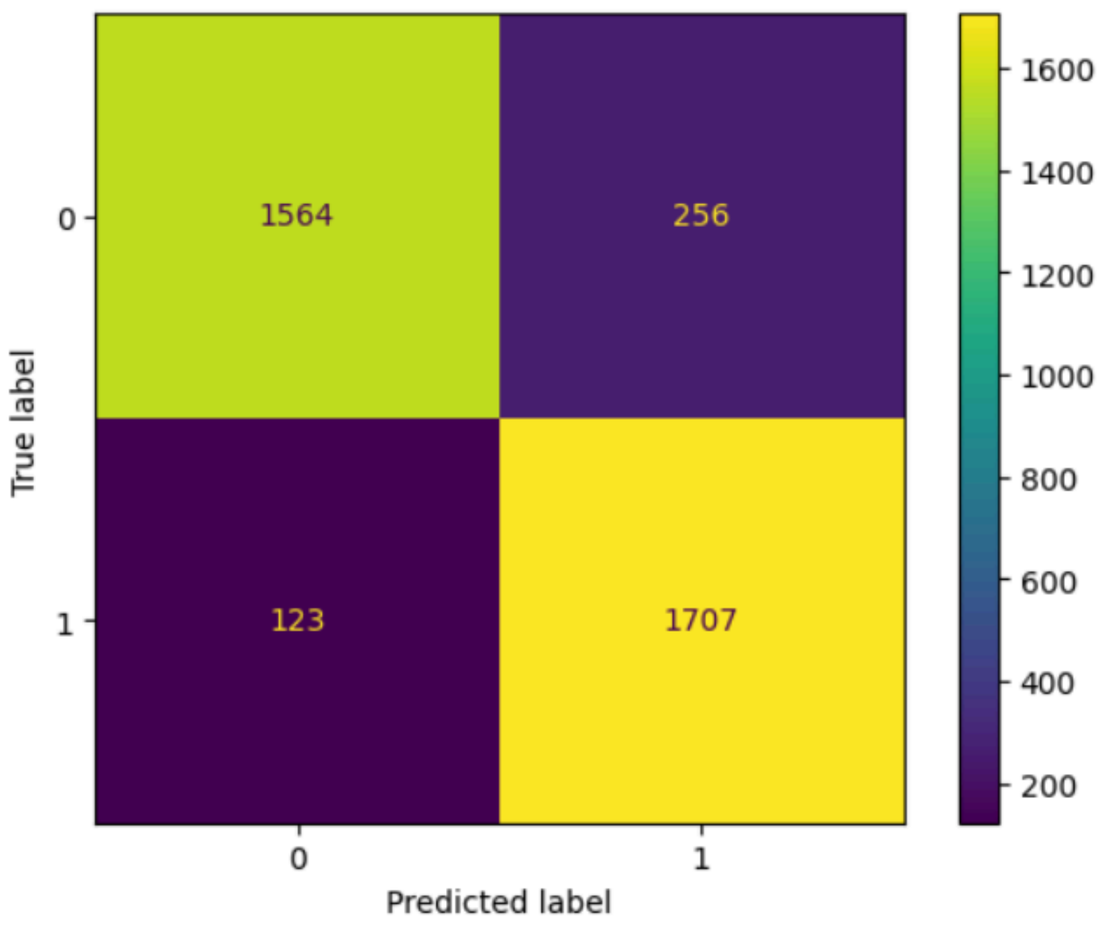
Plot Interpretation: The target points are overlapping and there are no obvious clusters. However, they capture and explain about 81% of the data.

Visualization Strategy

Testing the linear regression model on the test data: The classification report showed that The model did a good job predicting.

- 1564 records were correctly predicted as Conventional Avocado type.
- 123 records were correctly predicted as Conventional Avocado type
- 1707 records were correctly predicted as Organic Avocado type.
- 256 records incorrectly predicted as Organic Avocado type.

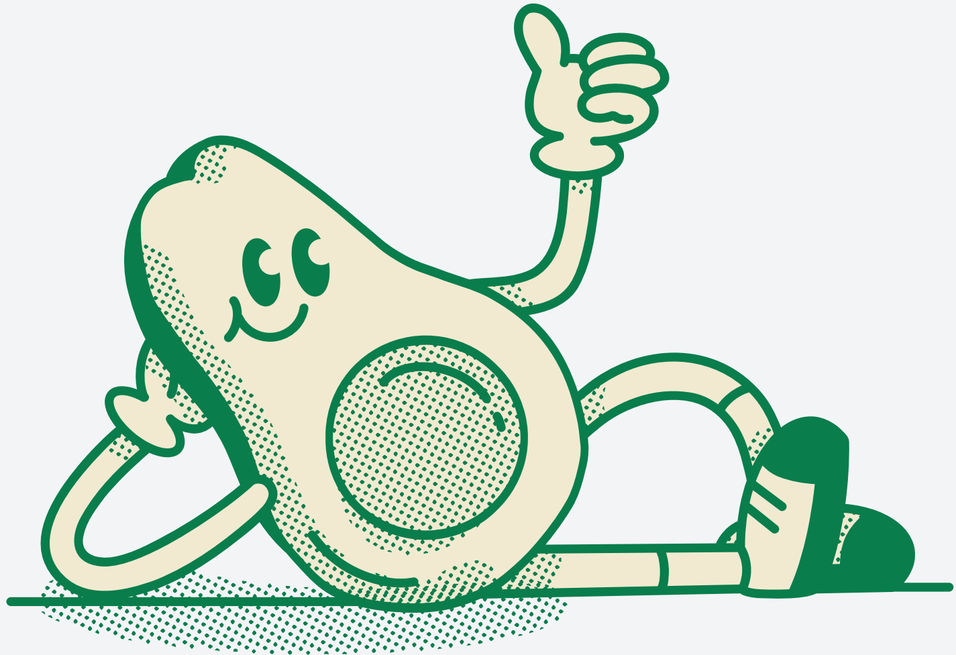
```
ConfusionMatrixDisplay.from_predictions(y_test, y_pred_pca)
plt.show()
```



The Confusion matrix displays the relationship between the predicted & true label. The model did a fair job predicting with an accuracy score of 90% which uis good enough.

Interpretation of Results

Results are documented on the notebook.



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Thank You

Questions are welcome

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